Teaching Teachers First and Then Student: Hierarchical Distillation to Improve Long-Tailed Object Recognition in Aerial Images

Wenda Zhao, *Member, IEEE*, Jiani Liu, Yu Liu, Fan Zhao, You He, and Huchuan Lu, *Senior Member, IEEE*

***Abstract*— Remote sensing data distribution generally exposes the long-tailed characteristic. This will limit the object recognition performance of existing deep models when they are trained with such unbalanced data. In this article, we propose a novel hierarchical distillation framework (HDF) to address the long-tailed object recognition in aerial images. First, we notice that not only student model should learn feature representations from teachers but also teacher models should learn feature representations from each other. Therefore, we build hierarchical teacher-wise distillation (HTWD) to improve the feature representations of the teacher models trained with middle and tail data, which is achieved by distilling the feature representations of the teacher model trained with head data. Second, we notice that the feature representations of the middle and tail classes cannot be effectively distilled from the teacher to the student since too little middle and tail data can be used to learn. Thus, we propose self-calibrated sampling (SCS) learning that enforces the student to strengthen the learning of the middle and tail data, thereby improving the student' feature learning ability. Extensive experiments on two widely used DOTA and FGSC-23 datasets demonstrate the superior performance of the proposed method compared with state-of-the-art methods. Model and code are publicly available at https://github.com/wdzhao123/T2FTS.**

***Index Terms*— Aerial images, hierarchical teacher-wise distillation (HTWD), long-tailed object recognition, self-calibrated sampling (SCS) learning.**

I. Introduction

DEEP networks have achieved great progress in object recognition, benefitting from large-scale and high-quality datasets, such as ImageNet [1], DOTA [2], and LFS [3]. However, data distribution in these datasets generally has the long-tailed characteristic (i.e., head classes occupy most of a dataset, while tail classes take up a small part of the dataset), especially for remote sensing datasets that are collected with professional labeling. Unfortunately, existing deep models for aerial images may not achieve the expected performances when they deal with such extremely unbalanced data [4]–[9]. Thus, how to improve long-tailed object recognition in aerial images is a valuable topic.

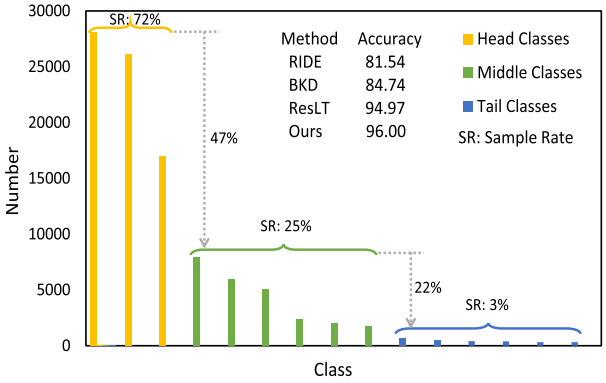


Fig. 1. Performance comparison of different methods on the long-tailed dataset DOTA [2]. DOTA is an aerial dataset that exposes the long-tailed characteristic. Our proposed method shows superior performance against state-of-the-art methods, RIDE [20], BKD [21], and ResLT [22].

Generally, there are two ways to tackle this issue: resampling methods [10]–[14] and cost-sensitive reweighting methods [15]–[19]. These methods improve performances by making models pay more attention to the tail classes. However, resampling methods may oversample tail classes or undersample head classes, which will lead models to underfit the heads or overfit the tails. Reweighting methods set larger weights for tail class loss, which may be sensitive to selected hyperparameters.

Recently, multiexpert networks are developed [20], [23]–[28]. They usually reorganize long-tailed data into groups and then train multiple expert models using different group data. After that, multiple expert models are integrated through simple mean strategy [24], [25], knowledge distillation [23], [26], or gating network [20], [27]. Multiexpert networks have shown great promise in the area of long-tailed tasks. However, multiple expert models

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may not pay attention to the relationship among experts. Besides, when multiple expert models are integrated, they ignore the long-tailed data issue.

In contrast to the small sample size problem, the long-tailed object recognition issue can take advantage of the large head data to improve the model performance on the small tail data. In this article, we propose a novel hierarchical distillation framework (HDF) to address the long-tailed object recognition in aerial images. HDF includes hierarchical teacher-wise distillation (HTWD) network and self-calibrated sampling (SCS) network. HTWD aims to improve the feature representations of middle and tailed data, and SCS focuses on distilling knowledge of middle and tailed data from teachers to student effectively.

Specifically, we first divide the long-tailed dataset into three groups (i.e., head class D_H, middle class D_M, and tail class D_T), according to their sample number. Then, three teacher models of $\mathcal{T}_H$, $\mathcal{T}_M$, and $\mathcal{T}_T$ are designed and trained with corresponding D_H, D_M, and D_T step by step. Since D_H has more samples than D_M and D_T, the corresponding teacher model $\mathcal{T}_H$ can produce better feature representations for each class in D_H. Sequentially, we distill the feature representation ability of $\mathcal{T}_H$ to $\mathcal{T}_M$ and $\mathcal{T}_T$ in the process of training $\mathcal{T}_M$ and $\mathcal{T}_T$ on the corresponding datasets D_M and D_T. Therefore, each teacher model will produce good feature representations with the help of its senior teachers. Finally, we implement a student network to learn feature representations from each teacher. However, if we directly distill feature representations of each teacher to student network, it cannot learn the tail features well due to a small number of tail samples. Therefore, we further propose SCS, which dynamically selects classes through the feature learning quality assessments. Thus, we eliminate the incomplete feature learning of tail classes by assigning them more sample training opportunities, thereby improving long-tailed object recognition in aerial images. As shown in Fig. 1, RIDE [20], BKD [21], and ResLT [22], which are part of the SoTA for long-tailed recognition on natural images, have limited performances on DOTA.

The main contributions are summarized as follows.

1) We propose a novel HDF to improve the long-tailed object recognition in aerial images, which is based on insight that not only student should learn feature representations from teachers but also teachers should learn feature representations from each other.
2) Teacher-wise distillation is built to improve the feature representations of teachers $\mathcal{T}_M$ and $\mathcal{T}_T$ by distilling the feature representation ability of $\mathcal{T}_H$ to $\mathcal{T}_M$ and $\mathcal{T}_T$ gradually. Moreover, SCS is introduced to improve the student's feature learning ability of tail data.
3) Extensive experiments on the two widely used DOTA and FGSC-23 datasets demonstrate the excellent performance of our method.

II. Related Work

A. Long-Tailed Object Recognition

Long-tailed object recognition is challenging due to the extremely imbalanced data distribution [29], [30]. Most methods address this problem using either resampling [10]–[14], [31]–[35] or cost-sensitive reweighting [15]–[19], [36]–[42]. Resampling methods can be generally divided into two types: 1) oversampling the tail classes [13], [14], [31], [32] and 2) undersampling the head classes [10]–[12]. For example, Ren *et al.* [32] proposed an ensemble-based adaptive oversampling algorithm to reduce the induction biases introduced from imbalanced data and overcome the class imbalance problem. Tahir *et al.* [11] proposed a novel inverse random undersampling method, which severely undersamples the majority class. Cost-sensitive reweighting methods usually assign large weights to the loss function for the tail class so that the loss function pays more attention to the tail class. For instance, Cui *et al.* [15] introduced a novel theoretical framework to measure data overlap by associating with each sample a small neighboring region and designed a reweighting scheme that uses the effective number of samples for each class to rebalance the loss. Ren *et al.* [17] proposed a novel meta-learning algorithm that learns to assign weights to training examples based on their gradient directions.

Oversampling methods repeatedly sample the tail data and train the same data multiple times, which will cause overfitting to the tail data. Undersampling methods ignore the head data, which may lead to underfitting of the head data. In addition, reweighting methods are sensitive to the choice of weights and will destroy the original structure of the data. In contrast, we propose multiteacher hierarchical distillation model to effectively utilize head data and tail data. Specifically, we first implement a teacher $\mathcal{T}_H$ model to learn feature representations of head data. Then, we distill the feature representation ability of $\mathcal{T}_H$ to another teacher $\mathcal{T}_T$ in the process of learning feature representations of tail data. Therefore, teacher $\mathcal{T}_T$ can learn feature representations well from tail data with the help of teacher $\mathcal{T}_H$.

B. Multiexpert Learning

Recently, multiexpert network learning methods have been developed for handling with the long-tailed problem [20], [23]–[28]. For example, Xiang *et al.* [23] introduced two levels of adaptive learning schedules: self-paced expert selection and curriculum instance selection, so that the knowledge is adaptively transferred to the student from three well-trained expert models. Zhou *et al.* [24] proposed a unified bilateral-branch network to take care of both representation learning and classifier learning, and then, the predicted outputs of these bilateral models are aggregated in the cumulative learning part. Wang *et al.* [20] implemented shared earlier layers and independent channel-reduced later layers to reduce the model bias with a distribution-aware diversity loss.

However, these methods do not pay attention to the connection between teacher models. In other words, the advantage of the good feature representations by the teacher trained with head data is not fully utilized. Therefore, we propose to teach teachers first and then student, where we distill the feature representation ability of the teacher trained with head data to the teacher trained with tail data. Thus, feature extraction ability of the teacher trained with tail data is improved.

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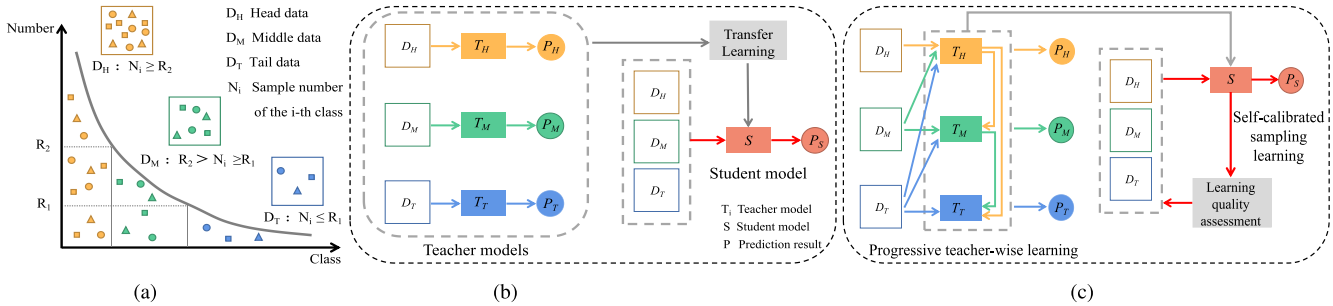


Fig. 2. Illustration of our motivation and framework. Multiexpert learning architecture may not fully utilize the good feature representations by the teacher T_H trained with head data D_H to help the teachers T_M and T_T trained with middle data D_M and tail data D_T, respectively. Besides, transfer learning from the teacher to the student ignores the small data sampling (e.g., middle and tail data). In contrast, our framework introduces hierarchical teacher-wise learning and self-calibrated sampling learning to address these two issues, helping the student model S get better feature representations, thereby improving the accuracy of long-tailed object recognition in aerial images. (a) Data subset division. (b) Multiexpert learning architecture. (c) Our framework.

C. Knowledge Distillation

Knowledge distillation [43] aims to transfer the knowledge learned from the teacher model to the student model. It has been widely used in vision tasks. For example, Chen *et al.* [44] proposed to learn compact and fast object detection networks using knowledge distillation. Xu *et al.* [45] exploited noisy self-supervision signals with selective transfer for distillation. Recently, knowledge distillation is used to handle the long-tailed problem [21], [46], [47]. He *et al.* [46] distilled the knowledge learned with virtual examples to tackle the long-tailed visual recognition problem. Ju *et al.* [47] divided the long-tailed data into multiple class subsets according to prior knowledge and enforce the model to learn the subset-specific knowledge.

We notice that the feature representation of the tail class cannot be effectively distilled from the teacher to the student since too little tail data can be used to learn. Different from the above methods, we propose an SCS learning mechanism to address this issue. Specifically, we evaluate the learning performance of the student model based on its forward propagation loss. The sample with large loss indicates that the student model cannot well learn the feature representation of the corresponding class, and we will assign a larger weight to increase the sampling probability of the class. Therefore, the student model is enforced to strengthen the learning of the small sample class.

III. Method

In this section, we first present the motivation and framework in detail, aiming to boost feature representations of teacher and student models in terms of long-tailed object recognition in aerial images. Then, we elaborate on two mechanisms of hierarchical teacher-wise learning and SCS learning. Finally, we give the optimizing detail of our method.

A. Motivation and Framework

Multiexpert networks have been developed for handling the long-tailed problem [20], [23]–[28]. Multiexpert learning generally takes the research route as follows. Given a long-tailed dataset D with C classes, K thresholds are set to divide D into $K+1$ subsets according to the number of samples per class. Here, we take K to 2 for example, and two thresholds R_1 and R_2 are selected based on the consideration that keeps the data in each subset as evenly distributed as possible. Thus, D is divided into three subsets of D_H, D_M, and D_T, as shown in Fig. 2(a). Then, multiexpert learning architecture is built, as shown in Fig. 2(b). Three teacher models $\mathcal{T}_H$, $\mathcal{T}_M$, and $\mathcal{T}_T$ are trained with the corresponding data subsets D_H, D_M, and D_T to learn feature representations, respectively. Sequentially, the knowledge learned from the teacher models is transferred to the student model.

However, existing multiexpert networks have two limitations.

1) The advantage of the good feature representations by the teacher trained with head data is not fully utilized to help the teacher trained with tail data.
2) Transfer learning from the teacher to the student ignores the small tail data such that the feature representations of the tail class cannot be effectively distilled.

Therefore, we propose hierarchical teacher-wise learning and SCS learning to address these two issues, helping the network get better feature representations thereby improving the accuracy of long-tailed object recognition in aerial images.

Our framework is shown in Fig. 2(c). On the one hand, we do not expect that the tail and middle class data to limit the learning ability of the teachers $\mathcal{T}_M$ and $\mathcal{T}_T$. Thus, we begin to implement the teacher model $\mathcal{T}_H$ to learn feature representations of head data. Then, we distill the feature representation ability of previous teachers to the current teacher. Therefore, teachers $\mathcal{T}_M$ and $\mathcal{T}_T$ can learn feature representations well from middle and tail data with the help of teacher T_H. On the other hand, the feature representations of the middle and tail classes may be biased toward the head class in the process of distilling the knowledge from the teacher to the student due to the small middle and tail data. Thus, we introduce learning quality assessment to guide the training sample sampling, thereby enforcing the student to strengthen the learning of the small sample class.

B. Hierarchical Teacher-Wise Learning

Since there is a huge gap of data number among subsets, the feature extraction ability of each teacher model trained with

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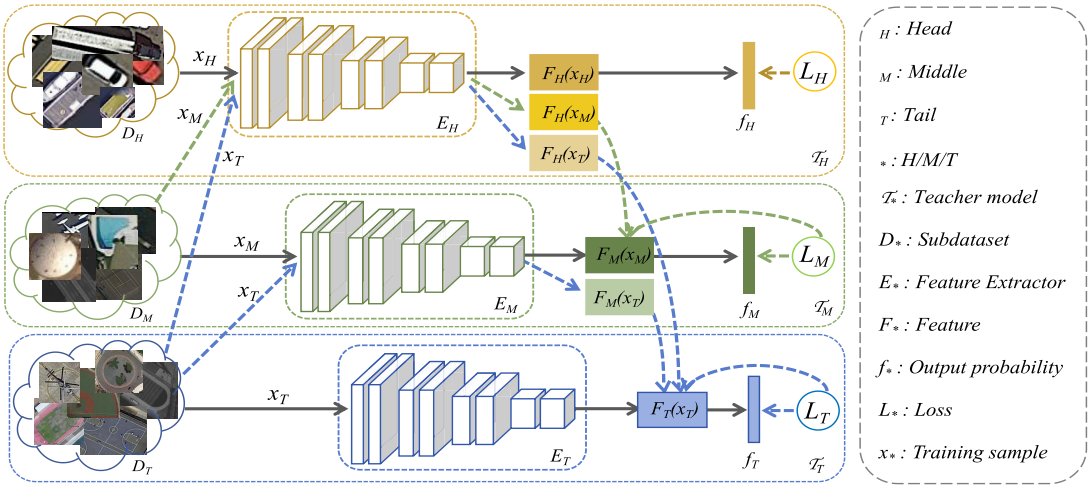


Fig. 3. Schema of hierarchical teacher-wise learning. We begin to train the teacher model $\mathcal{T}_H$ with D_H to learn feature representation well. Then, we distill the feature representation of $\mathcal{T}_H$ to $\mathcal{T}_M$ in the process of training $\mathcal{T}_M$ with D_M, where the training sample x_M is first fed to $\mathcal{T}_H$ to generate feature representation (denoted as green dashed line), and then, the feature representation is used to optimize $\mathcal{T}_M$ with the input x_M (denoted as solid line). Finally, we distill the feature representations of $\mathcal{T}_H$ and $\mathcal{T}_M$ to $\mathcal{T}_T$ in the process of training $\mathcal{T}_T$ with D_T, where the training sample x_T is first fed to $\mathcal{T}_H$ and $\mathcal{T}_M$ to obtain feature representations (denoted as blue dashed line), and then, the feature representations are adopted to optimize $\mathcal{T}_T$ with the input x_T (denoted as solid line).

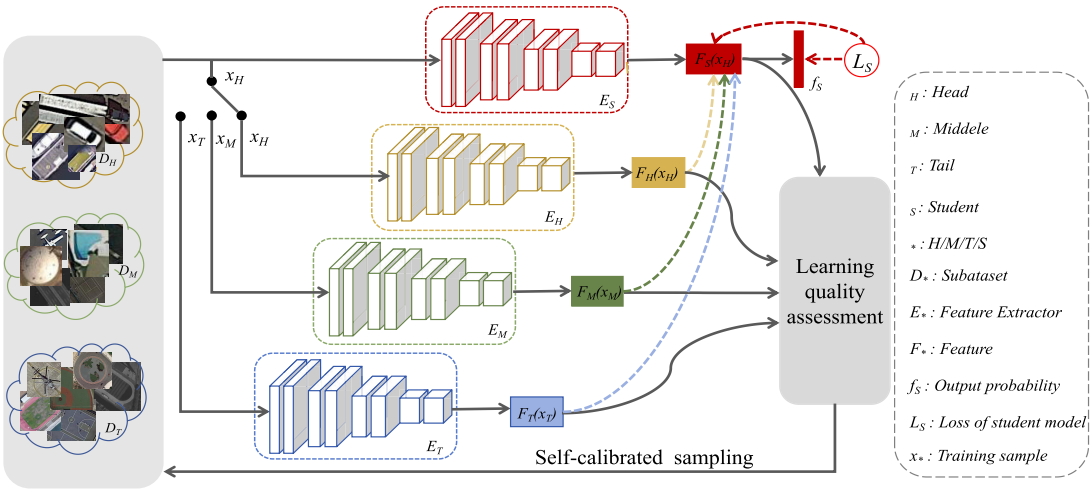


Fig. 4. Framework of self-calibrated sampling learning. Feature representations of the teacher models are distilled to the student model. In particular, learning quality assessment is built to automatically calibrate the sampling, where the class sampling probability increases to make the model be trained fully for the class that is difficult to learn.

the corresponding subset is uneven. In other words, the teacher model $\mathcal{T}_H$ trained with the head data D_H has the best feature extraction ability, followed by the teacher model $\mathcal{T}_M$ trained with the middle data D_M, and the teacher model $\mathcal{T}_T$ trained with the tail data D_T is the worst. Therefore, we propose hierarchical teacher-wise learning to take advantage of the good feature extraction capability of $\mathcal{T}_H$, helping $\mathcal{T}_M$ and $\mathcal{T}_T$ improve feature representation.

The schema of hierarchical teacher-wise learning is shown in Fig. 3. We first train T_H using D_H to make T_H gain good feature extraction capability, and the loss is

$$\mathcal{L}_H = -\frac{1}{N_H} \sum_{x_H \in D_H} P(x_H) \log(f_H(x_H)) \tag{1}$$

where x_H stands for a training sample in the head subset, N_H is the total number of training samples in the head subset, and $f_H(x_H)$ and $P(x_H)$ express the output probability of $\mathcal{T}_H$ and the ground truth of x_H, respectively.

Then, we train T_M with subset D_M. Since $\mathcal{T}_H$ has obtained good feature representation ability, we can utilize $\mathcal{T}_H$ to help $\mathcal{T}_M$ improve the learning performance. Therefore, we distill the feature representation of $\mathcal{T}_H$ to $\mathcal{T}_M$ in the process of training $\mathcal{T}_M$ with D_M. This can be written as

$$\mathcal{L}_M = -\frac{1}{N_M} \sum_{x_M \in D_M} P(x_M) \log(f_M(x_M)) + \frac{1}{N_M} \sum_{x_M \in D_M} (F_M(x_M) - F_H(x_M))^2 \tag{2}$$

where $F_M(x_M)$ and $F_H(x_M)$ are the features generated by $\mathcal{T}_M$ and $\mathcal{T}_H$ for the training sample x_M, respectively, N_M is the total number of training samples in the middle subset, and

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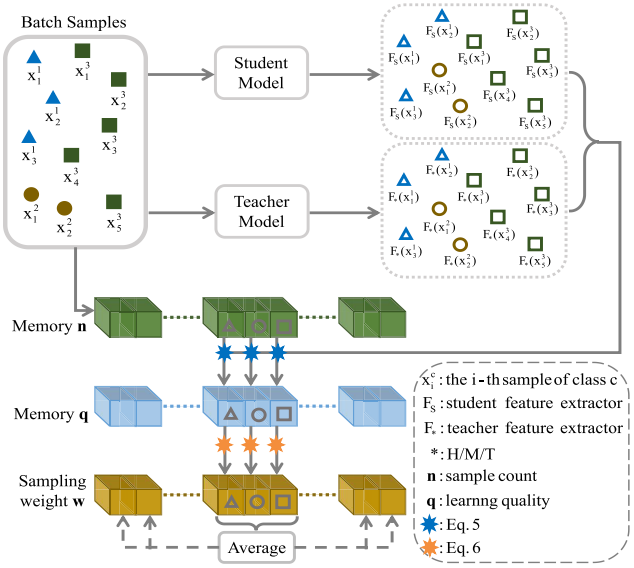


Fig. 5. Illustration of calculating sampling weight **w**. The learning quality of the student model for each class in a batch is first estimated. Then, the sampling weight **w** is calculated. The zero elements in **w** are assigned to the average sampling weight of the other nonzero elements.

$f_M(x_M)$ and $P(x_M)$ are the output probability of $\mathcal{T}_M$ and the ground truth of x_M, respectively.

Finally, we train T_T with subset D_T. In particular, we distill the feature representations of $\mathcal{T}_H$ and $\mathcal{T}_M$ to $\mathcal{T}_T$ in the process of training $\mathcal{T}_T$ with D_T, which can be defined as

$$\begin{aligned}\mathcal{L}_T = &-\frac{1}{N_T}\sum_{x_T\in D_T} P(x_T)\log(f_T(x_T)) \\ &+\frac{1}{N_T}\sum_{x_T\in D_T}(F_T(x_T)-F_H(x_T))^2 \\ &+\frac{1}{N_T}\sum_{x_T\in D_T}(F_T(x_T)-F_M(x_T))^2 \end{aligned} \tag{3}$$

where $F_T(x_T)$, $F_M(x_T)$, and $F_H(x_T)$ are the features generated by $\mathcal{T}_T$, $\mathcal{T}_M$, and $\mathcal{T}_H$ for the training sample x_T, respectively, N_T is the total number of training samples in the tail subset, and $f_T(x_T)$ and $P(x_T)$ stand for the output probability of $\mathcal{T}_T$ and ground truth of x_T, respectively. Thus, we take advantage of the good feature extraction capability of $\mathcal{T}_H$ to help $\mathcal{T}_M$ and $\mathcal{T}_T$ improve feature representation, which is achieved through hierarchical teacher-wise learning.

C. SCS Learning

Once we acquire the well-trained teacher models, we can distill their knowledge to the student model. Since the tail data is scarce, the knowledge learned from the tail data may not be completely distilled into the student model. However, existing methods (e.g., [23], [26]) usually ignore this issue. We address this problem from the perspective of sampling data. For the class c that is difficult to learn, we design a sampling weight w_c to increase the class sampling probability that makes the model be trained fully. Here, "sampling" means the selection of training images in batch size. Moreover, learning quality assessment is built to automatically calibrate the sampling weight. The framework is shown in Fig. 4.

Specifically, given a training image $x_i, i = H, M, or\ T$ sampled from $D = \{D_H, D_M, D_T\}$, we distill its feature representation $F_i(x_i)$ of the corresponding teacher model T_i to the student model S with the following loss:

$$\begin{aligned}\mathcal{L}_S = &-\frac{1}{N_B}\sum_{x_i\in \mathbf{w}D} P(x_i)\log(f_S(x_i)) \\ &+\frac{1}{N_B}\sum_{x_i\in \mathbf{w}D}(F_i(x_i)-F_S(x_i))^2 \end{aligned} \tag{4}$$

where $F_S(x_i)$ is the feature generated by S for the training image x_i, N_B stands for a batch size, and $f_S(x_i)$ and $P(x_i)$ express the output probability of S and class ground truth of x_i, respectively. $\mathbf{w} = \{w_1, w_2, \ldots, w_C\}$ stands for sampling weight for each class, and C is the total number of classes, which represents all the object classes in the dataset.

As shown in Fig. 5, given a memory $\mathbf{n} = [n_1, n_2, \ldots, n_C]$ as the sample count for each class in a batch and a memory $\mathbf{q} = [q_1, q_2, \ldots, q_C]$ as the learning quality of the student model for each class, we build the learning quality assessment to update sampling weight **w** for the next batch training. For a class $c \in C$, a larger difference between the features extracted by the student and teacher models indicates a worse learning quality of the student, and this class needs to be learned more. Thus, the learning quality assessment q_c for class c can be designed as follows:

$$q_c = \frac{1}{n_c}\sum_{i=1}^{n_c}\left(F_S\left(x_i^c\right)-F_i\left(x_i^c\right)\right)^2 \tag{5}$$

where x_i^c is a input sample of class c and F_i is the feature representation by corresponding teacher model (e.g., $\mathcal{T}_H$, $\mathcal{T}_M$, or $\mathcal{T}_T$). Then, the sampling weight w_c can be defined as

$$w_c = \alpha\log(q_c+1) \tag{6}$$

where α is a hyperparameter. Note that for the classes that are not sampled in the current batch, we assign them the average sample weight of the nonzero elements classes in the batch so that the zero-element classes can be sampled in the next batch.

Fig. 6 shows a visual comparison of our approach with SCS learning and without SCS learning. Our approach with SCS learning produces more uniform object features and less background noise than the one without SCS learning (see the red dashed boxes).

D. Optimizing

1) Structure: We use Resnet-50 [48] as our backbone network. Resnet-50 is composed of four residual blocks and followed by a fully connected operation to facilitate classification task. In our framework, we use the first three residual blocks as the feature extractor and the fourth residual block and fully connected operation as the classification head. Three teacher models and one student model have the same structure.

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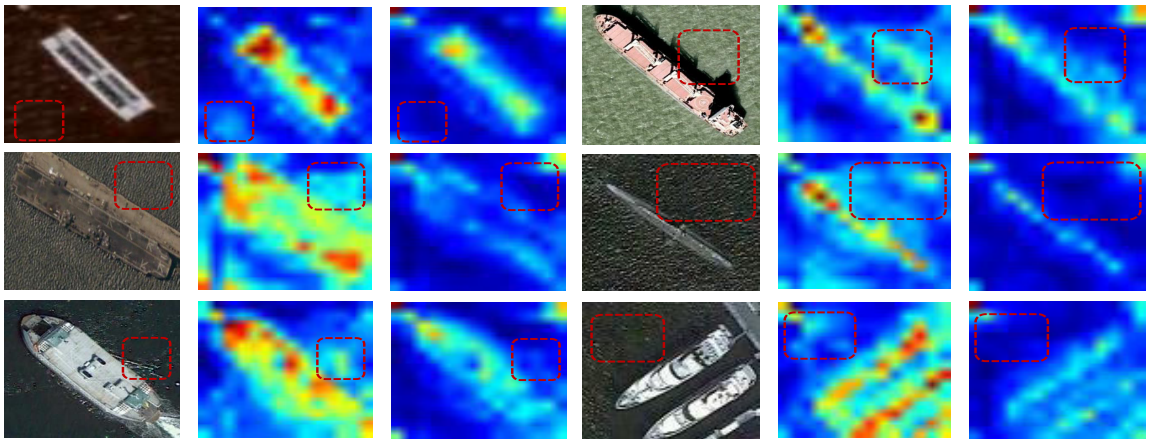


Fig. 6. Feature maps of our approach with (w) self-calibrated sampling learning (SCS) and without (w/o) SCS. The features are first extracted from the third block of Resnet-50 [48] and then are averaged to produce a feature map. The color from blue to red illustrates that the corresponding feature is lighted from weak to high. The sources and feature maps of the head samples, middle samples, and tail samples are from the first row to the third row. The features of the ships generated by Ours w SCS are more uniform and have less background noise than the ones generated by Ours w/o SCS. (a1) Source. (b1) Ours w/o SCS. (c1) Ours w SCS. (a2) Source. (b2) Ours w/o SCS. (c2) Ours w SCS.

TABLE I
STATISTICAL DETAILS OF DOTA DATASET

Subset	Head			Middle						Tail					
Class	Large-vehicle	Ship	Small-vehicle	Bridge	Harbor	Plane	Storage-tank	Swimming-pool	Tennis-court	Baseball-diamond	Basketball-court	Ground-track-field	Helicopter	Roundabout	Soccer-ball-field
Identification	TA1	TA2	TA3	TA4	TA5	TA6	TA7	TA8	TA9	TA10	TA11	TA12	TA13	TA14	TA15
Number	16969	28068	26126	2047	5983	7971	5029	1736	2367	415	515	325	630	399	326

TABLE II
STATISTICAL DETAILS OF FGSC-23 DATASET

Subset	Head					Middle						Tail										
Class	Non ship	Destoryer	Frigate	Cruiser	Bulk carrier	Aircraft carrier	Amphibious assult ship	Submarine	Combat boat	Auxiliary ship	Oil tanker	Landing craft	Amphibious transport dock	Tarawa class amphibious assult ship	Command ship	Medical ship	Container ship	Car carrier	Hover craft	Fishing boat	Passenger ship	Liquefied gas ship	Barge
Identification	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12	SC13	SC14	SC15	SC16	SC17	SC18	SC19	SC20	SC21	SC22	SC23
Number	387	434	236	234	274	132	123	190	114	180	132	86	72	70	71	17	81	58	96	82	70	74	43

2) Training: The overall training process includes two stages. In the first stage, we progressively train the three teacher models. Specifically, we first train T_H with the loss of L_H. Then, we fix T_H and train T_M with the loss of L_M. Finally, we fix T_H and T_M and train T_T using the loss of L_T. Now, we can get three teacher models with better feature

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TABLE III

EFFECTIVENESS STUDY OF HIERARCHICAL TEACHER-WISE DISTILLATION (HTWD) USING TOP-1 ACCURACY (%) ON FGSC-23. AVE_ACC STANDS FOR THE AVERAGE ACCURACY IN A SUBSET

	Head					Middle						Tail											
Setting	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12	SC13	SC14	SC15	SC16	SC17	SC18	SC19	SC20	SC21	SC22	SC23
Training $\mathcal{T}_H$, $\mathcal{T}_M$ and $\mathcal{T}_T$ jointly	90.72	87.04	91.53	**91.53**	79.71	**91.18**	**96.77**	93.75	93.10	62.22	66.67	81.82	83.33	**99.99**	72.22	99.99	80.00	**99.99**	**99.99**	65.00	66.67	**99.99**	**99.99**
	Ave_acc of $\mathcal{T}_H$: 88.01					Ave_acc of $\mathcal{T}_M$: 83.18						Ave_acc of $\mathcal{T}_T$: 86.38											
Training $\mathcal{T}_H$, $\mathcal{T}_M$ and $\mathcal{T}_T$ independently	**93.81**	**90.74**	**98.31**	88.14	**92.75**	**91.18**	**96.77**	95.83	**96.55**	88.89	**96.97**	**95.45**	83.33	94.44	**94.44**	**99.99**	**95.00**	**99.99**	**99.99**	**75.00**	77.78	95.00	**99.99**
	Ave_acc of $\mathcal{T}_H$: **92.60**					Ave_acc of $\mathcal{T}_M$: 94.09						Ave_acc of $\mathcal{T}_T$: 92.02											
Training $\mathcal{T}_H$, $\mathcal{T}_M$ and $\mathcal{T}_T$ with HTWD	**93.81**	**90.74**	**98.31**	88.14	**92.75**	**91.18**	**96.77**	**97.92**	89.66	**93.33**	**96.97**	90.91	**88.89**	**99.99**	**94.44**	**99.99**	**95.00**	**99.99**	**99.99**	**75.00**	**83.33**	95.00	**99.99**
	Ave_acc of $\mathcal{T}_H$: **92.60**					Ave_acc of $\mathcal{T}_M$: **94.55**						Ave_acc of $\mathcal{T}_T$: **92.96**											

TABLE IV

IMPORTANCE STUDY OF SELF-CALIBRATED SAMPLING LEARNING USING TOP-1 ACCURACY (%) OF THE STUDENT MODEL ON FGSC-23. AVE_ACC STANDS FOR THE AVERAGE ACCURACY IN A SUBSET

		Head					Middle						Tail											
Setting	Average	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12	SC13	SC14	SC15	SC16	SC17	SC18	SC19	SC20	SC21	SC22	SC23
Random sampling	88.00	88.66	89.81	**94.92**	88.14	**84.06**	88.24	96.77	95.83	**99.99**	75.56	84.85	77.27	72.22	94.44	83.33	**99.99**	**85.00**	**99.99**	**99.99**	**55.00**	66.67	95.00	**99.99**
		$Ave_acc:$ 89.03					$Ave_acc:$ 89.55						$Ave_acc:$ 84.51											
Class-aware sampling	88.12	**91.75**	90.74	**94.92**	86.44	81.16	88.24	**99.99**	95.83	96.55	73.33	87.88	**81.82**	**83.33**	94.44	83.33	**99.99**	**85.00**	92.86	**99.99**	40.00	66.67	**99.99**	**99.99**
		$Ave_acc:$ **89.29**					$Ave_acc:$ 89.55						$Ave_acc:$ 84.51											
Self-calibrated sampling	**88.73**	87.63	**91.67**	**94.92**	**89.83**	75.36	**91.18**	96.77	**99.99**	89.66	**82.22**	**90.91**	**81.82**	77.78	**99.99**	**94.44**	**99.99**	**85.00**	92.86	**99.99**	50.00	**77.78**	95.00	**99.99**
		$Ave_acc:$ 88.01					$Ave_acc:$ **91.82**						$Ave_acc:$ **86.85**											

representations for the head, middle, and tail data. In the second stage, we distill the knowledge learned by the teacher models to the student model by the loss of $\mathcal{L}_S$. In particular, sampling weight w is updated on each batch training with the proposed learning quality assessment.

Note that the proposed HTWD framework is based on the intuition that feature representations learned by a strong teacher trained on the head data could be distilled to weaker teachers trained on less frequent data to improve their performance. Therefore, we gradually train a set of "teacher" neural networks each specialized on separate set of classes, partitioned from the head to the tail data distribution.

IV. EXPERIMENTS

A. Experimental Settings

1) Datasets: We evaluate the proposed method on two public long-tailed datasets: DOTA [2] and FGSC-23 [50]. DOTA contains 15 categories and a total of 188 282 instances. In order to facilitate our experiment, we crop the instances and get a total of 98 906 training images and 28 853 testing images. The number of training images varies from 325 to 28 068 per class (see Table I for statistical details). FGSC-23 consists of 23 fine-grained categories of the ships and a total of 4081 images, where 3256 images belong to the training set and the rest 825 images are the testing set. The number of training images varies from 17 to 434 per class (see Table II for statistical details). The images are resized to 224 × 224. We comprehensively consider the number of images in each class and divide them into three subsets (i.e., head, middle, and tail subsets) with the thresholds $R_1 = 1000$ and $R_2 = 10\,000$ in DOTA and $R_1 = 100$ and $R_2 = 200$ in FGSC-23.

2) Implementation Details: We implement the experiments by PyTorch [51] on a PC with an RTX 2080 Ti GPU. The models are optimized using SGD with momentum 0.9, and the batch size is set to 8. The initial learning rate is 0.001 and multiplied by 10 every 50 epochs. α is taken to 75. Our model trained on FGSC costs about 7 h, where the teacher models converge to about 43 epochs on average and the student models converge to about 30 epochs. Our

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TABLE V

INFLUENCE STUDY OF HYPERPARAMETER α ON SELF-CALIBRATED SAMPLING USING TOP-1 ACCURACY (%) OF THE STUDENT MODEL ON FGSC-23. AVE_ACC STANDS FOR THE AVERAGE ACCURACY IN A SUBSET

		Head					Middle						Tail											
Setting	Average	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12	SC13	SC14	SC15	SC16	SC17	SC18	SC19	SC20	SC21	SC22	SC23
25	88.36	88.66	89.81	**96.61**	86.44	**84.06**	88.24	**99.99**	95.83	**96.55**	68.89	**93.94**	**86.36**	**77.78**	**99.99**	**94.44**	**99.99**	**85.00**	92.86	**99.99**	50.00	61.11	**99.99**	90.91
		$Ave_acc:$ **89.03**					$Ave_acc:$ 89.55						$Ave_acc:$ 85.92											
50	88.24	**89.69**	87.96	94.92	84.75	79.71	**94.12**	**99.99**	**99.99**	93.10	75.56	84.85	**86.36**	**77.78**	**99.99**	83.33	**99.99**	**85.00**	92.86	**99.99**	45.00	**83.33**	**99.99**	**99.99**
		$Ave_acc:$ 87.50					$Ave_acc:$ 90.91						$Ave_acc:$ **86.85**											
75	**88.73**	87.63	**91.67**	94.92	89.83	75.36	91.18	96.77	**99.99**	89.66	**82.22**	90.91	81.82	**77.78**	**99.99**	**94.44**	**99.99**	**85.00**	92.86	**99.99**	50.00	77.78	95.00	**99.99**
		$Ave_acc:$ 88.01					$Ave_acc:$ **91.82**						$Ave_acc:$ **86.85**											
100	87.52	**89.69**	90.74	91.53	**91.53**	81.16	91.18	93.55	95.83	93.10	62.22	84.85	77.27	**77.78**	**99.99**	83.33	**99.99**	**85.00**	**99.99**	95.83	**65.00**	66.67	**99.99**	**99.99**
		$Ave_acc:$ **89.03**					$Ave_acc:$ 85.91						$Ave_acc:$ 86.38											

TABLE VI

TOP-1 ACCURACY (%) OF DIFFERENT METHODS ON THE TESTING SET OF FGSC-23 DATASET. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED

		Head					Middle						Tail											
Method	Average	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12	SC13	SC14	SC15	SC16	SC17	SC18	SC19	SC20	SC21	SC22	SC23
RIDE [20]	52.27	73.21	33.33	18.31	28.99	26.76	82.81	35.37	55.56	72.83	35.71	49.41	26.15	39.73	31.71	33.33	60.66	55.88	82.35	89.33	46.97	44.12	94.64	97.06
BKD [21]	65.94	57.73	71.30	84.75	77.97	34.78	76.47	54.84	72.92	72.41	46.67	27.27	59.09	66.67	61.11	55.56	**99.99**	70.00	**99.99**	**99.99**	**65.00**	61.11	95.00	**99.99**
ResLT [22]	62.55	62.89	73.15	74.58	61.02	31.88	79.41	67.74	83.33	72.41	46.67	27.27	45.45	33.33	61.11	38.89	**99.99**	35.00	85.71	**99.99**	50.00	50.00	**99.99**	81.82
LDAM [40]	62.42	58.76	71.30	88.14	77.97	24.64	64.71	74.19	79.17	75.86	20.00	27.27	50.00	38.89	38.89	44.44	90.00	**85.00**	**99.99**	91.67	55.00	50.00	85.00	**99.99**
LAL [49]	53.45	75.26	72.22	42.37	18.64	47.83	64.71	58.06	68.75	79.31	42.22	21.21	27.27	27.78	27.78	5.56	80.00	50.00	42.86	91.67	40.00	44.44	45.00	**99.99**
T2FTS (ours)	**88.73**	**87.63**	**91.67**	**94.92**	**89.83**	**75.36**	**91.18**	**96.77**	**99.99**	**89.66**	**82.22**	**90.91**	**81.82**	**77.78**	**99.99**	**94.44**	**99.99**	**85.00**	92.86	**99.99**	50.00	**77.78**	95.00	**99.99**

model trained on DOTA costs about five days, where the teacher models converge to about 43 epochs on average and the student models converge to about 53 epochs. Our test model has parameters of 23.5M and FLOPs of 1.56G and takes about 6.8 ms to generate the result for an image with a size of 224 $\times$ 224.

Top-1 accuracy is used to evaluate the performance. Per-class accuracy is the accuracy on the subset of ground truth and predictions of a class.

B. Ablation Studies

1) Effectiveness of HTWD: In Section III-B, we propose an HTWD mechanism to improve the feature representations of middle data and tail data, thereby increasing the accuracy of corresponding object recognition. In order to verify the validity of HTWD, we compare our method (training $\mathcal{T}_H$, $\mathcal{T}_M$, and $\mathcal{T}_T$ with HTWD) with the method of training $\mathcal{T}_H$, $\mathcal{T}_M$ and $\mathcal{T}_T$ independently and training $\mathcal{T}_H$, $\mathcal{T}_M$, and $\mathcal{T}_T$ jointly. "Training $\mathcal{T}_H$, $\mathcal{T}_M$, and $\mathcal{T}_T$ jointly" refers to the training with vanilla cross-entropy loss (no distillation) and represents the baseline of standard training with no account for the long-tailed distribution. We show the top-1 accuracy of each class and average accuracy (Ave_acc) of each subset of classes for teacher models in Table III. Our method of training $\mathcal{T}_H$, $\mathcal{T}_M$, and $\mathcal{T}_T$ with HTWD improves the recognition accuracy generated by $\mathcal{T}_M$ and $\mathcal{T}_T$.

2) Importance of SCS Learning: Since the middle subset and the tail subset have smaller samples than the head subset, the knowledge of the teacher models $\mathcal{T}_M$ and $\mathcal{T}_T$ may not be completely distilled into the student model. Thus, the recognition accuracy of the student model will be limited on the middle subset and tail subset. In Section III-C, we propose an SCS learning mechanism that automatically calibrates the sampling weight to address this issue. In order to verify the importance of the SCS learning, we compare it with two sampling strategies of random sampling and class-aware

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TABLE VII

TOP-1 ACCURACY (%) OF DIFFERENT METHODS ON THE TESTING SET OF DOTA DATASET. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED

		Head			Middle						Tail					
Method	Average	TA1	TA2	TA3	TA4	TA5	TA6	TA7	TA8	TA9	TA10	TA11	TA12	TA13	TA14	TA15
RIDE [20]	81.54	78.72	81.12	87.29	92.66	84.19	90.74	88.51	89.25	90.20	73.64	76.51	61.24	**86.53**	69.07	73.29
BKD [21]	84.74	82.61	87.04	93.78	65.09	84.31	80.40	82.83	88.41	81.71	87.38	40.15	33.33	36.99	15.64	54.90
ResLT [22]	94.97	89.45	96.17	**95.50**	**97.57**	96.41	96.04	94.90	92.18	61.76	82.35	**91.30**	**90.48**	54.55	**82.14**	83.33
LDAM [40]	89.16	83.93	89.97	94.12	79.53	91.05	90.44	91.00	**92.27**	89.08	81.31	77.27	61.81	35.62	54.19	67.32
LAL [49]	92.77	92.04	96.83	94.10	84.05	91.53	96.52	90.58	83.86	84.74	83.18	50.76	74.31	45.21	51.40	55.56
T2FTS (ours)	**96.00**	**94.19**	**98.36**	94.58	93.75	**97.18**	**99.53**	**97.82**	66.59	**95.53**	**95.79**	87.12	88.89	79.45	79.89	**86.93**

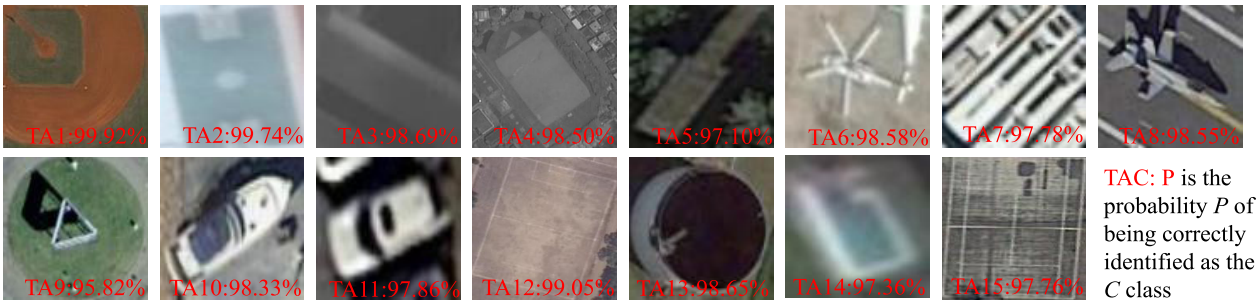


Fig. 7. Qualitative results of our method on the DOTA dataset.

TABLE VIII

PERFORMANCE OF OUR MODEL FOR NOISY AERIAL IMAGES ON THE TESTING SET OF FGSC-23 DATASET. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED

		Head					Middle						Tail										
Method	Average	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12	SC13	SC14	SC15	SC16	SC17	SC18	SC19	SC20	SC21	SC22	SC23
Ours(noise free)	88.73	87.63	91.67	94.92	89.83	75.36	91.18	96.77	99.99	89.66	82.22	90.91	81.82	77.78	99.99	94.44	99.99	85.00	92.86	99.99	50.00	77.78	95.00	99.99
Ours(noise)	87.15	86.60	89.81	94.92	89.83	75.36	91.18	96.77	97.92	89.66	80.00	90.91	77.27	77.78	94.44	94.44	99.99	85.00	92.86	99.99	30.00	66.67	95.00	99.99

sampling [14]. Results of different sampling strategies are shown in Table IV. Our SCS learning mechanism achieves the best overall accuracy of 88.73%. Moreover, Ave_acc values of our method on the middle subset and tail subset are better than both random sampling and class-aware sampling.

3) Influence of Hyperparameter α on SCS: We build a learning quality assessment to update sampling weight w_c for the class c in (6) in the process of batch-wise training, where a hyperparameter α is introduced to adjust the influence of w_c. A larger α will produce a bigger w. To obtain a suitable α, we conduct a series of experiments by taking α to 25, 50, 75, and 100. The top-1 accuracy of each class and the average accuracy (Ave_acc) of each subset classes for the student model are shown in Table V. We can see that the results are better when α is taken to 75, which can achieve the overall Ave_acc 88.73%, and Ave_acc values of middle subset and tail subset of 91.82% and 86.85%, respectively.

C. Comparison With State of the Art

In this section, we compare our proposed method (T2FTS) with other state-of-the-art methods, including RIDE [20], BKD [21], ResLT [22], LDAM [40], and LAL [49]. RIDE and ResLT use ResNet50 as the backbone, and BKD, LDAM, and LAL use ResNet32 as the backbone. For fair comparisons, we utilize the authors' original codes and fine-tune them with the corresponding training dataset.

1) Comparison on FGSC-23: The results of different methods on the testing set of FGSC-23 dataset are reported in

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TABLE IX

PERFORMANCE OF OUR MODEL FOR NOISY AERIAL IMAGES ON THE TESTING SET OF DOTA DATASET. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED

		Head			Middle						Tail					
Method	Average	TA1	TA2	TA3	TA4	TA5	TA6	TA7	TA8	TA9	TA10	TA11	TA12	TA13	TA14	TA15
Ours(noise free)	96.00	94.19	98.36	94.58	93.75	97.18	99.53	97.82	66.59	95.53	95.79	87.12	88.89	79.45	79.89	86.93
Ours(noise)	93.84	94.46	97.56	93.66	73.92	97.08	96.13	91.41	65.68	94.21	71.03	82.58	70.14	80.82	49.16	90.85

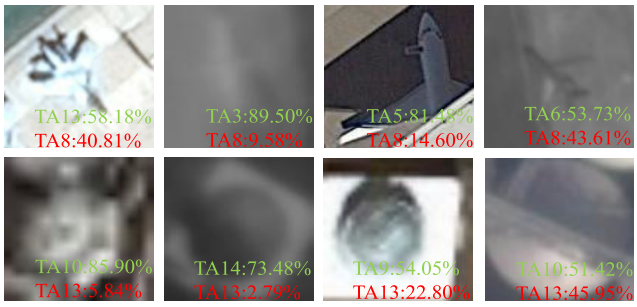


Fig. 8. Failure examples of our method on the DOTA dataset. Green text stands for being mistakenly identified and corresponding probability. Red text is correctly identified and corresponding probability.



Fig. 9. Visual examples of noisy aerial images.

Table VI. We can see that our method has the highest average accuracy of 88.73%, which is 20%–30% higher than the other methods. Especially for the middle and tail classes, our method generally achieves superior performance compared with state-of-the-art methods.

2) Comparison on DOTA: Table VII shows the results of different methods on the testing set of DOTA. Our method also has the best effect with the average accuracy of 96%. In general, our method has improved classification accuracy for the head, middle, and tail classes. Therefore, our method can improve the classification performances of the middle and tail classes, thereby addressing the impact of the long-tailed object recognition in aerial images. In particular, the class of TA8 has lower accuracy than the ones of other methods. The reason is that the objects (swimming-pool) of TA8 are generally low resolution, as shown in the first row of Fig. 8.

3) Discussion: The proposed HTWD framework utilizes the strong teacher trained on the head data to help weaker teachers trained on less frequent data improve their performance. In addition, our SCS improves the distillation on the tail data. Therefore, our method achieves higher performance compared to the state of the art. Qualitative results of our method are shown in Fig. 7. The class is correctly identified in a large probability. However, our model is limited for low-resolution object with less details, as shown in Fig. 8. Future work will address this, e.g., study image super-resolution.

D. Noise Robustness

Here, we provide the object recognition performance of the proposed model for noisy aerial images. Noisy images are generated by adding Gaussian noise with random variance that is sampled between 0.0005 and 0.001, as shown in Fig. 9. In Tables VIII and IX, we can see that our model for noisy aerial images [Ours(noise)] achieves competitive object recognition performance compared with our model for noise-free aerial images [Ours(noise free)].

V. CONCLUSION

We study the long-tailed object recognition problem in aerial images and present a novel HDF to improve the recognition performance of the middle and tail data. In particular, we propose the HTWD that distills the feature representations of the teacher model trained with head data to the teacher models trained with middle and tail data, thereby improving the feature representations of the middle and tail data. Moreover, in order to effectively distill the feature representations of the middle and tail classes from the teachers to the student, we conduct the SCS learning that enforces the student to strengthen the learning of the middle and tail data, thereby improving the student' feature learning ability. Extensive experimental results illustrate that the proposed method achieves superior performance than other state-of-the-art methods.

REFERENCES

[1] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[2] G.-S. Xia *et al.*, "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[3] W. Zhao *et al.*, "Feature balance for fine-grained object classification in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:00 UTC from IEEE Xplore. Restrictions apply.

[4] Q. Liu, X. Xiang, Z. Yang, Y. Hu, and Y. Hong, "Arbitrary direction ship detection in remote-sensing images based on multitask learning and multiregion feature fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1553–1564, Feb. 2021.

[5] T. Zhang *et al.*, "HOG-ShipCLSNet: A novel deep learning network with hog feature fusion for SAR ship classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5210322.

[6] J. Shi, Z. Jiang, and H. Zhang, "Few-shot ship classification in optical remote sensing images using nearest neighbor prototype representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3581–3590, 2021.

[7] L. Zeng *et al.*, "Dual-polarized SAR ship grained classification based on CNN with hybrid channel feature loss," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[8] J. He, Y. Wang, and H. Liu, "Ship classification in medium-resolution SAR images via densely connected triplet CNNs integrating Fisher discrimination regularized metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3022–3039, Apr. 2021.

[9] Y. Xu and H. Lang, "Ship classification in SAR images with geometric transfer metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6799–6813, Aug. 2021.

[10] P. Jeatrakul, K. W. Wong, and C. C. Fung, "Classification of imbalanced data by combining the complementary neural network and smote algorithm," in *Proc. Int. Conf. Neural Inf. Process.*, Cham, Switzerland: Springer, 2010, pp. 152–159.

[11] M. A. Tahir, J. Kittler, and F. Yan, "Inverse random under sampling for class imbalance problem and its application to multi-label classification," *Pattern Recognit.*, vol. 45, no. 10, pp. 3738–3750, 2012.

[12] M. Buda, A. Maki, and M. A. Mazurowski, "A systematic study of the class imbalance problem in convolutional neural networks," *Neural Netw.*, vol. 106, pp. 249–259, Oct. 2018.

[13] J. Byrd and Z. Lipton, "What is the effect of importance weighting in deep learning?" in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 872–881.

[14] L. Shen, Z. Lin, and Q. Huang, "Relay backpropagation for effective learning of deep convolutional neural networks," in *Proc. Eur. Conf. Comput. Vis.*, Cham, Switzerland: Springer, 2016, pp. 467–482.

[15] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, "Class-balanced loss based on effective number of samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9268–9277.

[16] C. Huang, Y. Li, C. C. Loy, and X. Tang, "Learning deep representation for imbalanced classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 5375–5384.

[17] M. Ren, W. Zeng, B. Yang, and R. Urtasun, "Learning to reweight examples for robust deep learning," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 4334–4343.

[18] Y.-X. Wang, D. Ramanan, and M. Hebert, "Learning to model the tail," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 7032–7042.

[19] J. Tan *et al.*, "Equalization loss for long-tailed object recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11662–11671.

[20] X. Wang, L. Lian, Z. Miao, Z. Liu, and S. X. Yu, "Long-tailed recognition by routing diverse distribution-aware experts," 2020, *arXiv:2010.01809*.

[21] S. Zhang, C. Chen, X. Hu, and S. Peng, "Balanced knowledge distillation for long-tailed learning," 2021, *arXiv:2104.10510*.

[22] J. Cui, S. Liu, Z. Tian, Z. Zhong, and J. Jia, "ResLT: Residual learning for long-tailed recognition," 2021, *arXiv:2101.10633*.

[23] L. Xiang, G. Ding, and J. Han, "Learning from multiple experts: Self-paced knowledge distillation for long-tailed classification," in *Proc. Eur. Conf. Comput. Vis.*, Cham, Switzerland: Springer, 2020, pp. 247–263.

[24] B. Zhou, Q. Cui, X.-S. Wei, and Z.-M. Chen, "BBN: Bilateral-branch network with cumulative learning for long-tailed visual recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9719–9728.

[25] P. Wang, K. Han, X.-S. Wei, L. Zhang, and L. Wang, "Contrastive learning based hybrid networks for long-tailed image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 943–952.

[26] A. Iscen, A. Araujo, B. Gong, and C. Schmid, "Class-balanced distillation for long-tailed visual recognition," 2021, *arXiv:2104.05279*.

[27] Y. Zhao, W. Chen, X. Tan, K. Huang, and J. Zhu, "Adaptive logit adjustment loss for long-tailed visual recognition," 2021, *arXiv:2104.06094*.

[28] S. Sharma, N. Yu, M. Fritz, and B. Schiele, "Long-tailed recognition using class-balanced experts," in *Proc. DAGM German Conf. Pattern Recognit.*, Tübingen, Germany: Springer, Sep. 2021, pp. 86–100.

[29] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263–1284, Jun. 2009.

[30] W. Ouyang, X. Wang, C. Zhang, and X. Yang, "Factors in finetuning deep model for object detection with long-tail distribution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 864–873.

[31] S. Ando and C. Y. Huang, "Deep over-sampling framework for classifying imbalanced data," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discovery Databases*. Cham, Switzerland: Springer, 2017, pp. 770–785.

[32] F. Ren, P. Cao, W. Li, D. Zhao, and O. Zaiane, "Ensemble based adaptive over-sampling method for imbalanced data learning in computer aided detection of microaneurysm," *Comput. Med. Imag. Graph.*, vol. 55, pp. 54–67, Jan. 2017.

[33] B. Kang *et al.*, "Decoupling representation and classifier for long-tailed recognition," 2019, *arXiv:1910.09217*.

[34] G. Haixiang *et al.*, "Learning from class-imbalanced data: Review of methods and applications," *Expert Syst. Appl.*, vol. 73, pp. 220–239, May 2017.

[35] W. Yu, T. Yang, and C. Chen, "Towards resolving the challenge of long-tail distribution in UAV images for object detection," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 3258–3267.

[36] J. Ren *et al.*, "Balanced meta-softmax for long-tailed visual recognition," 2020, *arXiv:2007.10740*.

[37] C. Huang, Y. Li, C. C. Loy, and X. Tang, "Deep imbalanced learning for face recognition and attribute prediction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 11, pp. 2781–2794, Nov. 2020.

[38] S. H. Khan, M. Hayat, M. Bennamoun, F. A. Sohel, and R. Togneri, "Cost-sensitive learning of deep feature representations from imbalanced data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 8, pp. 3573–3587, Feb. 2017.

[39] M. A. Jamal, M. Brown, M.-H. Yang, L. Wang, and B. Gong, "Rethinking class-balanced methods for long-tailed visual recognition from a domain adaptation perspective," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 7610–7619.

[40] K. Cao, C. Wei, A. Gaidon, N. Arechiga, and T. Ma, "Learning imbalanced datasets with label-distribution-aware margin loss," 2019, *arXiv:1906.07413*.

[41] T. Wu, Q. Huang, Z. Liu, Y. Wang, and D. Lin, "Distribution-balanced loss for multi-label classification in long-tailed datasets," in *Proc. Eur. Conf. Comput. Vis.*, Cham, Switzerland: Springer, 2020, pp. 162–178.

[42] H. Yu, N. Zhang, S. Deng, Z. Yuan, Y. Jia, and H. Chen, "The devil is the classifier: Investigating long tail relation classification with decoupling analysis," 2020, *arXiv:2009.07022*.

[43] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.

[44] G. Chen, W. Choi, X. Yu, T. Han, and M. Chandraker, "Learning efficient object detection models with knowledge distillation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 742–751.

[45] G. Xu, Z. Liu, X. Li, and C. C. Loy, "Knowledge distillation meets self-supervision," in *Proc. Eur. Conf. Comput. Vis.*, Cham, Switzerland: Springer, 2020, pp. 588–604.

[46] Y.-Y. He, J. Wu, and X.-S. Wei, "Distilling virtual examples for long-tailed recognition," 2021, *arXiv:2103.15042*.

[47] L. Ju *et al.*, "Relational subsets knowledge distillation for long-tailed retinal diseases recognition," 2021, *arXiv:2104.11057*.

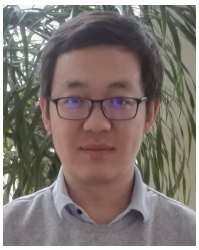
[48] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[49] A. K. Menon, S. Jayasumana, A. S. Rawat, H. Jain, A. Veit, and S. Kumar, "Long-tail learning via logit adjustment," 2020, *arXiv:2007.07314*.

[50] X. Zhang, Y. Lv, L. Yao, W. Xiong, and C. Fu, "A new benchmark and an attribute-guided multilevel feature representation network for fine-grained ship classification in optical remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1271–1285, 2020.

[51] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 8024–8035.

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